

Cameron Lee

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Education

Brown University — Master of Science, Computer Science
San Diego State University — Bachelor of Science, Computer Science

Starting Fall 2026
May 2026 | GPA: 3.7

Skills

Programming: C/C++, C#, Java, Python, Assembly, HTML/CSS/JavaScript, SQL, Bash
Platforms/Tools: Linux, Windows, macOS, Git, ROS 2, Vim, VSCode, Visual Studio, Unity

Experience

Undergraduate Research Assistant | CARE AI Lab **March 2026 - Present**

- Assemble and configure multiple PiCar-X robots to support repeatable experimentation in autonomous control.
- Investigate methods for the autonomous adjustment of PID control parameters to improve stability, responsiveness, and navigation performance on small robotic platforms.
- Support research across wheeled robots and the Unitree Go1 quadruped, with a focus on adaptive control and autonomous robotic behavior.

Instructional Assistant & Course Developer | SDSU CS Department **August 2025 - Present**

- Design, revise, and maintain programming assignments in collaboration with faculty for the first three undergraduate programming courses, helping modernize core coursework for SDSU's CS curriculum.
- Modernize assignment delivery, automated grading, and feedback workflows for legacy coursework, reducing manual overhead and improving course scalability.
- Collect student performance data and provide office hours, tutoring, and debugging support to guide course revisions and improve student outcomes.

Undergraduate Research Assistant | Computer System Intelligence Lab **June 2025 - August 2025**

- Conducted research on methods for quantifying uncertainty in Visual-Inertial Simultaneous Localization and Mapping (VISLAM) systems.
- Analyzed how hardware differences and computational environments impact uncertainty estimates and localization accuracy.

Manager | Abbey's Real Texas BBQ **March 2024 - January 2026**

- Promoted from cashier/server to manager, taking responsibility for supervising staff, coordinating daily operations, and maintaining service quality in a fast-paced environment.
 - Designed and implemented a standardized inventory system with checklists and par levels, reducing ordering time by ~50% and virtually eliminating midweek stockouts.
 - Improved scheduling and operational workflows to increase team efficiency and support more consistent day-to-day execution.
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Projects

FITENTH Autonomous Racecar | Senior Capstone, Real-Time Embedded Control Lab **Fall 2025 - Present**

- Developed and tuned LiDAR-based reactive planning and control algorithms for a 1/10-scale autonomous racecar using ROS 2, focusing on disparity-extension obstacle avoidance and Ackermann steering.
- Improved performance across open straights and narrow environments by iterating on speed-governing PD control, steering smoothing, and gap-selection behavior to reduce oscillation and support stable high-speed navigation.
- Integrated and validated on both simulated and physical hardware in a 5-member team, with work continuing beyond the capstone toward a workshop paper.

Linux Kernel Real-Time Scheduling Extensions | Theory of Real Time Systems **Fall 2025**

- Modified the Linux 4.9 kernel on a Raspberry Pi 3 to support explicit real-time task reservations, periodic execution, and end-to-end latency monitoring.
- Implemented custom kernel system calls and extended task_struct, scheduler, and timer logic to support Rate Monotonic and partitioned Earliest Deadline First scheduling models.
- Added per-task execution-time accounting across context switches and automated the cross-compilation, deployment, and testing workflow to accelerate kernel debugging and iteration.

Coda | Personal Project **Spring 2026 - Present**

- Develop a command-line application in C backed by SQLite to manage and search a personal music collection.
- Implement database initialization, record creation, editing, listing, and deletion with a flag based CLI interface for album metadata.
- Structure the project as a working, documented codebase with support for multiple database files and configurable storage paths.